

## Conceptual Checks

Fill each blank with the most precise physics term or relation. Use the sign conventions implied by the statement.

1. The electric field is defined as \_\_\_\_\_ per unit \_\_\_\_\_.
2. In electrostatic equilibrium, the electric field inside the conducting material is \_\_\_\_\_.
3. Electric flux through a closed surface depends on the \_\_\_\_\_ charge, not the charge outside.
4. A right-moving wave can be written  $y(x,t)=f(\text{_____})$ .
5. In SHM, acceleration is proportional to displacement and points in the \_\_\_\_\_ direction.
6. Refraction changes wave \_\_\_\_\_ but not source frequency.
7. Polarization  $P$  is dipole moment per unit \_\_\_\_\_.
8. Sound in air is primarily a \_\_\_\_\_ wave of compressions and rarefactions.
9. Force from a potential-energy graph is the \_\_\_\_\_ derivative of  $U$  with respect to  $r$ .
10. The common ion formed by chlorine is \_\_\_\_\_ because chlorine tends to \_\_\_\_\_ one electron.

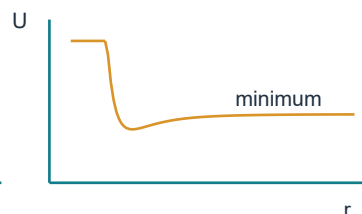
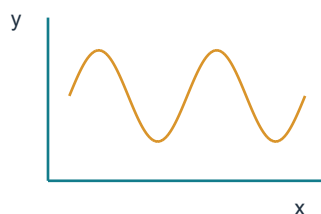
Short explanation: choose one blank and explain the physical reason for your answer.

## Matching: Visuals to Equations

Match each visual description (A-D) to the equation that best represents it (1-4). Write the letter-number pairs and explain one match.

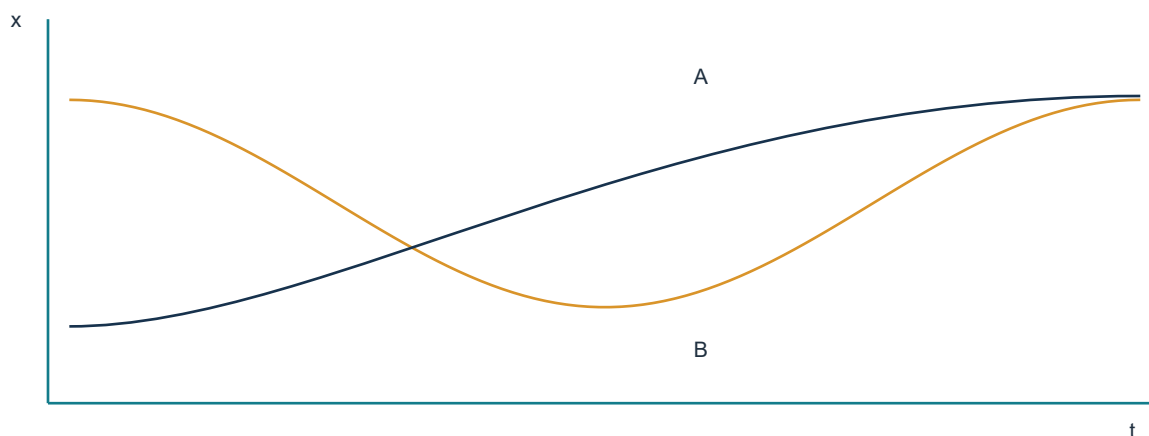
- A. A sphere centered on a point charge; E is constant on the surface.
- B. A sinusoidal pattern translating right without changing shape.
- C. A U(r) curve with a minimum at the stable separation.
- D. A dielectric filled with aligned microscopic dipoles.

1.  $\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}}/\epsilon_0$
2.  $y(x,t) = f(x-vt)$
3.  $F(r) = -dU/dr$
4.  $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$



## Simple Harmonic Motion

3. A mass obeys  $x(t) = A \cos(\omega t + \phi)$ . (a) Draw  $x$  versus  $t$  for  $A = 0.20$  m,  $\omega = 4.00$  rad/s,  $\phi = 0$ . (b) State the period, maximum speed, and acceleration when  $x = +A$ . (c) Of the two graphs below, identify which could represent velocity and justify the phase relationship.

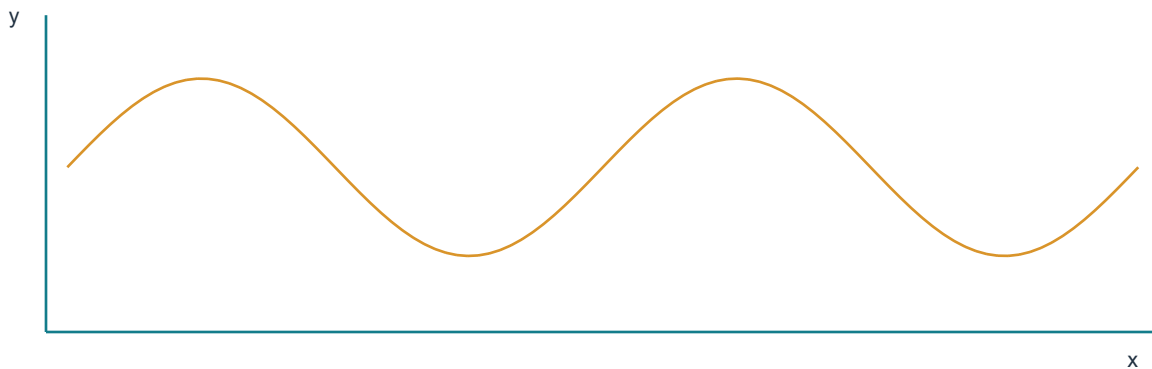


Graph A starts at  $+A$ ; Graph B starts at zero and rises. Circle the graph that can represent  $v(t)$  when  $x(t) = A \cos(\omega t)$ .

Show the energy relation you would use to find speed at a general position  $x$ .

## Traveling Wave: Multiple Quantities

4. A wave is  $y(x,t)=0.040 \cos(6.00x-120t)$  m, with  $x$  in metres and  $t$  in seconds. Find (a) amplitude, (b) wavelength, (c) frequency, (d) period, (e) propagation speed and direction, and (f) the transverse velocity at  $x=0, t=0$ .



State which parts come from the coefficient,  $k$ , and  $\omega$  before calculating.

$$k=2\pi/\lambda ; \omega=2\pi f ; v=\omega/k ; v_y=dy/dt$$

## Calculus Derivations: Constant Acceleration

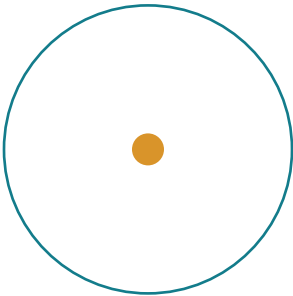
5. Starting from  $a(t)=dv/dt$ , use definite integration to prove  $v_f=v_i+at$  for constant  $a$ . Then derive  $x(t)=x_i+v_i t+1/2at^2$  from  $dx/dt=v(t)$ . State the initial conditions used in each integration and explain why an integration constant appears.

Required structure: differential statement  $\rightarrow$  integral limits  $\rightarrow$  evaluation  $\rightarrow$  physical interpretation.

$$a(t)=dv/dt ; v(t)=dx/dt$$

## Gauss's Law: Proof and Application

6. (a) Starting with  $E=kQ/r^2$  for a point charge, derive Gauss's law for a spherical surface and identify where  $k=1/(4\pi\epsilon_0)$  is used. (b) Apply Gauss's law to find  $E(r)$  outside and inside a uniformly charged solid sphere of radius  $R$  and density  $\rho$ . (c) Explain why the same shortcut does not work for an arbitrary nonsymmetric charge shape.



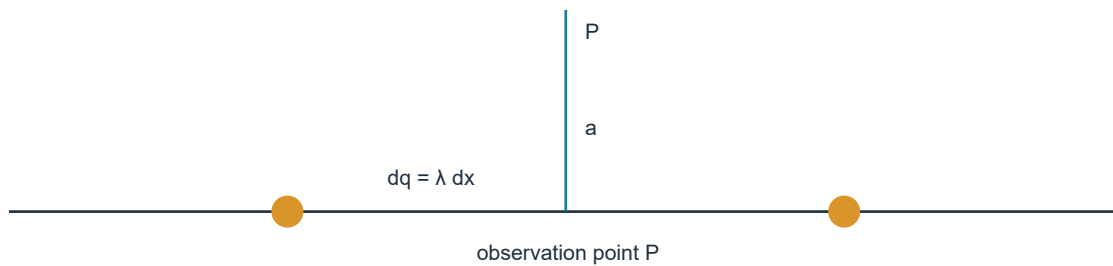
closed Gaussian surface

Show the surface area, enclosed charge, and region assumptions explicitly.

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} ; Q_{\text{enc}} = \rho(4\pi r^3/3)$$

## Continuous Charge Distribution and Trigonometric Substitution

7. A uniformly charged rod of length  $2L$  lies along the  $x$ -axis from  $-L$  to  $+L$  with linear density  $\lambda$ . Point  $P$  is on the  $y$ -axis a distance  $a$  above its midpoint. (a) Explain why horizontal field components cancel. (b) Starting from  $dE = k dq/r^2$ , use  $dq = \lambda dx$  and a trigonometric substitution to derive the exact field magnitude at  $P$ . (c) Check the limit  $L \rightarrow \infty$ .

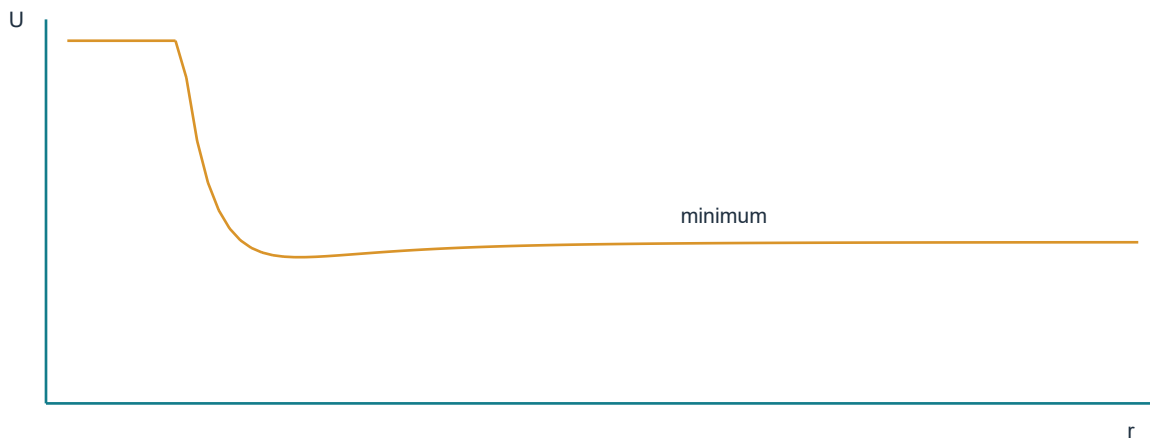


Show a substitution such as  $x = a \tan \theta$  (or an equivalent choice), transform  $dx$  and  $r$ , update the limits, and resolve the  $y$ -component using  $\cos$  or  $\sin$ .

$$dE_y = dE \cos \theta ; r = \sqrt{a^2 + x^2}$$

## Potential Energy and Graph Interpretation

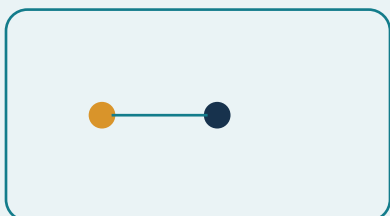
8. Consider  $U(r) = C_{12}/r^{12} - C_6/r^6$ . (a) Identify the graph below that best matches this expression and explain each region. (b) Derive  $F(r)$  by differentiating. (c) State the equilibrium condition and whether the minimum is stable. (d) Compare this model with  $U = kq_1q_2/r$ .



Include the physical meaning of sign, slope, and the  $r \rightarrow \infty$  limit.

## Atoms, Ions, and Dielectrics

9. (a) Explain the difference between the Bohr picture and the electron-cloud model. (b) Use valence-electron reasoning to predict  $\text{Na}^+$  and  $\text{Cl}^-$ . (c) For a linear isotropic dielectric, derive  $D = \epsilon E$  from  $p = qd$ ,  $P = (1/V)\sum p_i$ ,  $D = \epsilon_0 E + P$ , and  $P = \epsilon_0 \chi_e E$ . (d) Draw aligned dipoles and label  $E$ ,  $P$ , and  $D$ .

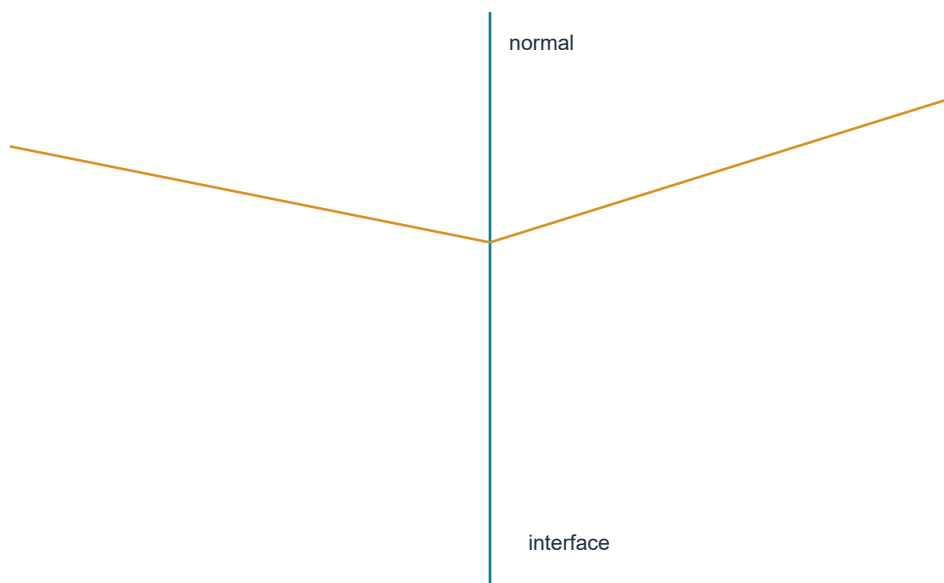


$E, P, D \rightarrow$

Use the diagram area for your response and label which quantity is the material response.

## Sound and Refraction

10. A sound has  $p_{\text{rms},1}=20\text{ }\mu\text{Pa}$  at one location and  $p_{\text{rms},2}=60\text{ }\mu\text{Pa}$  at another. (a) Find the change in sound-pressure level. (b) Explain why the pressure ratio is squared when converted through intensity. (c) A wave travels from a medium with  $n_1=1.00$  into  $n_2=1.50$  at  $\theta_1=40^\circ$ . Find  $\theta_2$  and state whether the ray bends toward or away from the normal. Draw the interface, normal, incident ray, and refracted ray.



Use  $L_p=20 \log_{10}(p_2/p_1)$  for a level difference and  $n_1 \sin \theta_1 = n_2 \sin \theta_2$  for refraction.

## Derivation Synthesis

11. Prove each relation from its differential or geometric starting point: (a) from  $F=-kx$  and  $F=mx''$ , derive the SHM differential equation; (b) from  $y=A \cos(kx-\omega t)$ , derive the 1-D wave equation and identify  $v$ ; (c) from equal phase travel along an interface, derive Snell law. For each, state the physical assumption that makes the derivation valid.

Use separate lines for the physical law, substitution, differentiation or geometry, and final relation.

$$x'' + \omega^2 x = 0 ; y_{xx} = (1/v^2) y_{tt} ; n_1 \sin \theta_1 = n_2 \sin \theta_2$$

## Point Charges: 1D and 2D Force and Field

12. Use  $k=8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$ . (a) On the x-axis,  $q_1=+3.0 \text{ } \mu\text{C}$  is at  $x=0$  and  $q_2=-2.0 \text{ } \mu\text{C}$  is at  $x=0.40 \text{ m}$ . Find the magnitude and direction of the force on  $q_2$ . (b) A test charge  $q_0=+1.0 \text{ } \mu\text{C}$  is at  $(0,0)$ ,  $q_1=+4.0 \text{ } \mu\text{C}$  at  $(0.30,0) \text{ m}$ , and  $q_2=+4.0 \text{ } \mu\text{C}$  at  $(0,0.30) \text{ m}$ . Find the x- and y-components of the net electric field at  $q_0$ , then its magnitude and direction. (c) Explain how the force result differs from the field result conceptually.

For full credit: draw displacement vectors, identify attraction/repulsion, write each vector contribution, resolve components, and report units. Leave the lower half of the page for work.

$$\mathbf{F}=k|q_1q_2|/r^2 ; \mathbf{E}=k|q|/r^2 ; E_x=\Sigma E_{i,x} ; E_y=\Sigma E_{i,y}$$

## Hallmark Equations and Physical Meaning

13. (a) Write the standard forms of Newton's universal law of gravitation, Coulomb's law, the electric-field definition, Gauss's law, the potential-energy relation, the linear wave equation, and the Doppler relation for a stationary observer and moving source. (b) For each, name the physical quantity being predicted and one condition or interpretation that prevents misuse. (c) Applications: predict qualitatively whether perceived frequency increases or decreases when the observer moves toward a sound source; predict how the field of an infinite line changes when distance doubles; and explain why a conductor surface cannot support a tangential electrostatic field.

Memory bank to complete from understanding, not guessing:

$F_g = \underline{\hspace{1cm}}$      $F_C = \underline{\hspace{1cm}}$      $E = \underline{\hspace{1cm}}$      $\Phi_E = \underline{\hspace{1cm}}$   
 $U = \underline{\hspace{1cm}}$      $y_{xx} = \underline{\hspace{1cm}}$      $f_{\text{obs}} = \underline{\hspace{1cm}}$

Use the bottom half to explain the three applications in complete sentences and include one direction sketch.